

Ricardo (Zhicong) Xie

Mechanical Engineering Intern | CAD, Revit MEP, HVAC, FEA/CFD and Project Engineering

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Valid NZ student visa | No sponsorship required | 25h/week during semester | Full-time 17 Nov 2026 - late Feb 2027

PROFILE

- Mechanical engineering master's student focused on CAD, Revit MEP, HVAC, FEA/CFD and practical project delivery.
- Portfolio includes NZ MVHR/ERV, IoT enclosure design, Abaqus composite FEA, CFD validation and workshop evidence.
- Project screenshots and source links: <https://zxie588-alt.github.io>

EDUCATION

University of Auckland - Master of Engineering Studies, Mechanical Engineering | 2026 - Present

Current study and project work in building services, smart manufacturing, CFD/thermal modelling, composite structures and engineering methods. Relevant coursework: Building Services Engineering, Advanced Industry 4.0 Smart Manufacturing, Manufacturing and Industrial Systems, Research Project

Huazhong Agricultural University - Bachelor of Engineering, Mechanical Engineering / Mechatronics | 2021 - 2025

Mechanical design, CAD, electrotechnics, sensors, embedded control and mechatronic system integration.

PRACTICAL AND TEAM EXPERIENCE

- Manufacturing workshop training: prepared AutoCAD/DXF laser-cut profiles and compared drawings with cut parts.
- Engineering practice: interpreted force-displacement tests, controlled a robotic arm and led a small automotive-structure walkthrough.
- Technical communication: former student news lead with photography and project-documentation experience.

PROJECT WORK

Small Residential MVHR / ERV Revit MEP Coordination

Revit 2026 MEP coordination, MVHR/ERV layout, airflow schedule, SolidWorks, AutoCAD/DXF, NZBC G4/H1 context, commissioning notes

- Built a NZ residential MVHR/ERV package with Revit MEP, CAD evidence and review notes.
- Modelled MVHR unit, ducts, terminals and hangers in SolidWorks; exported STEP, STL, DXF and PNG.
- Created a Revit 2026 model with plan, 3D view, airflow schedule, sheets and simple clash notes.
- Checked 60 L/s supply, 45 L/s extract and 85 L/s boost against duct velocity and pressure allowances.
- Added M-102 pressure review, BIM issue log and Mitsubishi Lossnay LGH-35RVX3-E selection note.
- Linked the design to NZBC G4/H1, Healthy Homes context and a commissioning test plan.

Ruggedized Outdoor IoT Sensor Module Design Package

SolidWorks, enclosure design, DFM notes, Keil C, validation workflow

- Designed a SolidWorks enclosure with gasket stack-up, screw bosses, PCB envelope and service access.
- Prepared DFM notes for CNC review, 3D printing and assembly tolerance checks.
- Screened corner-load, strap-lug and screw-boss regions with an FEA-style check.
- Built Keil C firmware structure with sensor, battery and state-machine modules.

Manufacturing Workshop Training & DXF Preparation

Laser cutting, AutoCAD/DXF, CNC machining exposure, 3D printing workflow

- Prepared AutoCAD/DXF geometry for laser cutting and checked the cut plate against drawing intent.
- Practised CNC workflow: workholding, tool motion, cutting fluid, chip formation and shop safety.
- Reviewed machined gears and shafts for tolerance, edge quality and tool-access constraints.

Composite Formula SAE Front Wing Analysis

Abaqus/Standard, shell FEA, composite layup, structural checks

- Built an Abaqus shell FEA model with S4R elements, laminate layups and local mesh refinement.
- Converted aero, lateral cornering and 4.1 G bump conditions into structural load cases.
- Checked ply-level stress use; peak utilisation was about 74% under the 4.1 G bump case.

Thermal-Fluid Simulation & Validation - Solar Chimney Case Study

ANSYS Fluent/CFX, conjugate heat transfer, natural convection

- Built a CFD validation workflow for a laboratory-scale solar chimney with coupled air, absorber and cover domains.
- Checked mesh sensitivity, mass/heat balance, Reynolds/Rayleigh scaling and solver behaviour.
- Documented model boundaries, limitations and recommended physical data points needed for future experimental correlation.

TECHNICAL SKILLS

- Design & Manufacturing: native Revit 2026 MEP coordination, SolidWorks assemblies, AutoCAD/DXF, DFM, drawing-to-part thinking, CNC workflow, 3D printing workflow, STEP/STL export
- Simulation & Analysis: Abaqus/Standard FEA, ANSYS Fluent/CFX CFD, composite shell analysis, thermal-fluid modelling, mesh and balance checks
- Systems & Testing: BIM coordination handoff, commissioning workflow, NZBC G4/H1 awareness, sensor integration, verification and validation, requirements and risk tracking